

B. GURUSAIPRASADREDDY

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EDUCATION

• Amrita Vishwa Vidyapeetham, Coimbatore

B.Tech in Computer Science and Engineering

2023 – 2027

CGPA: 6.76/10

TECHNICAL SKILLS AND INTERESTS

Languages: Java, Python, C

Frontend: HTML5, CSS3, JavaScript, React.js

Backend: Node.js, Express.js, FastAPI, RESTful API Design

Databases: MongoDB, MySQL

Developer Tools: Git, GitHub, VS Code, Linux, Eclipse, Docker, Wireshark, Cisco Packet Tracer

Technologies: MERN Stack (MongoDB, Express.js, React.js, Node.js)

Soft Skills: Problem Solving, Teamwork, Communication

Coursework: DSA, DBMS, OOPS Concept, OS, ML, Computer Networks, Software Engineering

PERSONAL PROJECTS

• Campus Bites (Canteen Management) | React, FastAPI, MongoDB, Socket.IO

2026

Real-time food ordering and vendor platform | [GitHub](#)

- Engineered a full-stack campus food ordering platform utilizing React.js, FastAPI, and MongoDB, featuring role-based dashboards to streamline menu management, digital checkout, and financial analytics.
- Architected a real-time tracking system with Socket.IO and WebSockets, delivering instant order status updates and live cart synchronization.
- Implemented an intelligent meal recommendation engine powered by Google GenAI, analyzing purchase history to drive user personalization.

• Automated Timetable System | MERN Stack, Python

2026

Constraint-based academic scheduling platform | [GitHub](#)

- Developed a full-stack platform using the MERN stack with secure JWT authentication and role-based access dashboards for Admins, Faculty, and Students.
- Integrated a Python constraint-based scheduling engine with the Node.js backend to automatically generate conflict-free timetables based on faculty workload and room availability.
- Architected RESTful APIs and MongoDB schemas to manage academic data, time slots, and leave requests, ensuring seamless data flow across the system.

• Embedded Defense Radar System | STM32, Keil IDE

June 2025

Prototype radar-based obstacle detection system | [GitHub](#)

- Developed a real-time radar simulation system on STM32 ARM Cortex-M4 using bare-metal Embedded C with interrupt-driven architecture.
- Implemented ultrasonic pulse-echo sensing and PWM-controlled servo scanning for continuous semi-circular object detection.
- Built low-level I2C drivers to interface with OLED display, enabling real-time radar visualization with dynamic detection indicators.

• Hotel Booking Demand Analysis | Python, Pandas, NumPy, Matplotlib, Seaborn

2026

Exploratory Data Analysis (EDA) | [GitHub](#)

- Performed comprehensive exploratory data analysis on hotel booking data to identify guest behavior, booking patterns, pricing trends, and cancellation drivers.
- Conducted data cleaning, missing value treatment, and feature engineering by creating derived metrics such as total nights and total guests.
- Built visualizations to analyze ADR trends, seasonal demand, market segments, meal preferences, guest origins, and hotel-wise cancellation patterns.
- Generated business insights through comparative analysis of City and Resort hotels, supporting data-driven decision making in the hospitality domain.

ACHIEVEMENTS

• 160 Days DSA Problem Solving Challenge (Certification) GeeksforGeeks

2024

• Finalist – NeuroHACK, Indian Institute of Technology (IIT) Madras

Secured Rank 5 in Round 1
Advanced to the final round of a national-level hackathon.

2025